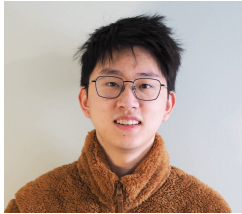


Wentao Lu

Msc in Software Engineering and Intelligent Systems
University of Alberta



CONTACT

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Edmonton, AB, Canada

SKILLS

- Python
- C/C++
- Java
- Go
- SQL
- Git
- Matlab
- Tensorflow
- Pytorch
- HTML

HARDWARE

- MSP 430
- STM 32
- EFR32FG12
- Xilinx Zynq Z7
- Assembly RISC-V

AWARDS&HONOR

- Alberta Graduate Excellence Scholarship
- First Class Standing of University of Alberta
- University of Alberta Maple Leaf First Year Excellence Scholarship
- International Student Scholarship of University of Alberta

SOCIALS



Hello there! I am a Master of Science student in Software Engineering and Intelligent Systems at the University of Alberta

Currently concluded my Master of Science in Software Engineering and Intelligent Systems at the University of Alberta, I have a solid foundation background in Computer Engineering from the same institution. As an active member of the ASGAARD lab, I am deeply involved in research focusing on Machine Learning System Engineering and Analytics. My academic and practical experiences have equipped me with a strong foundation in innovative technologies, preparing me to tackle complex challenges in the field

PROFESSIONAL EXPERIENCE

Software Development Engineer | [Electronic Arts](#)

Apr 2026 - Present

Quality Assurance and AI researches.

Science Researcher | [Electronic Arts](#)

Oct 2024 - Oct 2025

Quality Assurance and AI researcher applying AI/ML to improve large-scale game testing workflows and reduce manual QA effort through automation. Developed tools for automated bug detection and retrieval from gameplay footage using vision-language models and collaborated with QA and engineering teams to identify and deploy scalable solutions.

Graduate Research Assistant Fellowship | [University of Alberta](#)

Sept 2023 - Jan 2026

As a Graduate Research Assistant Fellow at the ASGAARD Lab, I engage in advanced research focused on Machine Learning System Engineering and Analytics. My role involves both individual and collaborative projects, aiming to innovate and refine methodologies within intelligent systems and ML/AI testing.

Undergraduate/Graduate Teaching Assistant | [University of Alberta](#)

Sept 2022 - Dec 2025

In charge of lab sessions for CPUT 291 and ECE 325. I guide students in mastering database management concepts with a focus on SQL, SQL in Python, and MongoDB in Python as well as introduction to Java. My role involves hands-on teaching to bridge theory with practical application, empowering students with critical skills for data manipulation and database design.

PUBLISHED WORK

How Far Can VLMs Go for Visual Bug Detection? Studying 19,738 Keyframes from 41 Hours of Gameplay Videos | [FSE 2026](#)

[Wentao Lu](#), [Alexander Senchenko](#), [Alan Sayle](#), [Abram Hindle](#), [Cor-Paul Bezemer](#) (Collaboration with Electronic Arts)

Automated Bug Frame Retrieval from Gameplay Videos Using Multimodal Large Language Models | [SEIP 2026](#)

[Wentao Lu](#), [Alexander Senchenko](#), [Abram Hindle](#), [Cor-Paul Bezemer](#) (Collaboration with Electronic Arts)

Analyzing Developer Use of ChatGPT Generated Code in Open Source GitHub Projects | [MSR 2024](#)

[Balreet Grewal](#), [Wentao Lu](#), [Sarah Nadi](#), [Cor-Paul Bezemer](#)

EDUCATION

Electrical and Computer Engineering, MSc | [University of Alberta](#)

Sept 2023 - Jan 2026

- Specialized in Intelligent System and Analytics
- CGPA 4.0/4.0

Computer Engineering, BSc | [University of Alberta](#)

Sept 2019 - June 2023

- Graduation with Distinction CGPA 3.7/4.0
- First Class Standing

PROJECTS

Quantum Permutation Pad Testbed | [Capstone project](#)

Jan 2023 - May 2023

- Design a testbed capable of encryption and decryption using an open-source classical implementation of the Quantum Permutation Pad (QPP) Algorithm on a system-on-chip (SoC) platform.
- Utilization of the FPGA fabric for implementation of the QPP and AES comparable to AES-256. Both algorithms are implemented using C++.
- Used on experiments with attacks, brute-force and man in the middle attack. Results of study of QPP as the cryptography standard in oppose to the emerging quantum computers.

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Space X Falcon9 first stage Landing Prediction | [IBM data science capstone](#)

Jun 2022 - Jul 2022

- Data collection using Web Scrapping BeautifulSoup and Space X API.
- Exploratory Data Analysis including data wangling, SQL queries, data visualization matplotlib/seaborn and interactive visual analytics Folium and Plotly Dash.
- Scikit learn Machine Learning Algorithms including KNN, Decision Tree, Logistic Regression and Support Vector Machine to perform prediction of Falcon9 landing.

Database management program using MongoDB | [Database management project](#)

Mar 2022 - Apr 2022

- Processing unregulated CSV files to JSON files to load in MongoDB sever.
- Using mongoDB client to implement a database management system for enormous quantity of data.

Unix system Concurrent Jobs simulation | [Operating system design project](#)

Mar 2022 - Apr 2022

- Utilize threads to simulate the concurrent execution of jobs by C/C++ under unix system. The system has number of resource types, and each resource type has several available units. All these resources' units are non-sharable and non-preemptable.

Peer process communication using FIFO/TCP | [Operating system design project](#)

Mar 2022 - Apr 2022

- Communicating peer processes/switches that utilize signals for examining the progress of the running processes, FIFOs/TCP for communication, and I/O multiplexing for nonblocking I/O.
- Mimic processes by designing Master switch/process and Packet switches/processes in C++.
- Packet switches read data and match the rule in its forwarding table for post processing.
- TOR Packet switches send requests to master switch (if no rules found locally) through FIFO or TCP connections over internet by developing client-server model.
- Master switch decides the request send to other packet switches or.
- (Local) Peer packet switches communication through FIFO.

Python Database management program using SQL | [Database management project](#)

Jan 2022 - Feb 2022

- Using SQL inside python to design and implement a database for streaming.
- Including login options for both customer and editor.

Unix Shell program | [Operating system design project](#)

Jan 2022 - Feb 2022

- Using C++ to design and mimic a Shell program like Bash.
- Implementing major commands using Unix system calls.